SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Questions

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA0515**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks50** **Duration :60 Minutes**

Annexure A

Scenario-Based Questions

Code of Conduct:

I R.Dinesh -192421117 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R Dinesh

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Marks (100)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario	
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues	
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts	
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided	
Clarity of Response	15%	Response is clear, structured, and	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand	

		logically presented				

Scenario-Based Questions

1. Object Boundary Detection

In a medical image, boundaries between tissues need to be clearly identified.

(a) Interpret the scenario and explain the importance of boundary detection.

(b) Identify key challenges in detecting boundaries.

(c) Apply a suitable edge detection method.

(d) Justify your selection with logical reasoning.

(e) Present your explanation clearly and systematically.

2. Simple Segmentation

An image contains objects clearly distinguishable from the background based on intensity.

(a) Interpret the scenario and explain segmentation requirements.

(b) Identify key features enabling separation.

(c) Apply a suitable segmentation technique.

(d) Justify your decision logically.

(e) Present your response clearly.

3. Complex Segmentation

An image contains multiple regions with overlapping intensity values.

(a) Interpret the scenario and explain segmentation challenges.

(b) Identify key issues such as intensity overlap.

(c) Apply an appropriate advanced segmentation method.

(d) Justify how the method handles complexity.

(e) Provide a structured and clear explanation.

4. Broken Edges

After applying edge detection, object boundaries appear discontinuous.

(a) Interpret the scenario and explain why edges are broken.

(b) Identify key issues affecting boundary continuity.

(c) Apply a suitable method to improve edge connectivity.

(d) Justify your approach logically.

(e) Present your explanation clearly.

5. Combined Requirement

An industrial inspection system requires noise removal while preserving important edges for defect detection.

(a) Interpret the scenario and define the combined requirement.

(b) Identify key issues involving noise and edge preservation.

(c) Apply an appropriate sequence of techniques.

(d) Justify the order and selection with strong reasoning.

(e) Present your response in a clear and structured format.

1. Object Boundary Detection

(a) Interpret the scenario and explain the importance of boundary detection.

In a medical image, different tissues such as bones, muscles, and organs have different boundaries. Boundary detection is used to identify the edges between these tissues. It helps doctors detect tumors, organs, and abnormal regions accurately.

(b) Identify key challenges in detecting boundaries.

- Presence of image noise
- Low contrast between tissues
- Blurred or weak edges
- Complex tissue structures

(c) Apply a suitable edge detection method.

Method: Canny Edge Detection

(d) Justify your selection with logical reasoning.

Canny edge detection is suitable because:

- It removes noise using Gaussian filtering.
- It detects both strong and weak edges.
- It produces thin and continuous edges.
- It has fewer false edge detections than other methods.

(e) Present your explanation clearly and systematically.

Flow:

1. Read the medical image.
2. Apply Gaussian filter to remove noise.
3. Apply Canny edge detector.
4. Obtain clear tissue boundaries.

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2. Simple Segmentation

(a) Interpret the scenario and explain segmentation requirements.

The objects are clearly different from the background based on intensity. The goal is to separate the object from the background.

(b) Identify key features enabling separation.

- High contrast
- Different intensity values
- Clear object boundaries

(c) Apply a suitable segmentation technique.

Method: Threshold Segmentation

(d) Justify your decision logically.

Threshold segmentation is suitable because:

- It is simple and fast.
- It separates pixels based on intensity.
- It works well when object and background have distinct intensity values.

(e) Present your response clearly.

Steps:

1. Read image.
2. Select threshold value.
3. Convert image into binary image.
4. Extract object from background.

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3. Complex Segmentation

(a) Interpret the scenario and explain segmentation challenges.

The image contains several objects whose intensity values overlap. Simple thresholding cannot separate them correctly.

(b) Identify key issues such as intensity overlap.

- Overlapping intensity values
- Uneven illumination
- Noise
- Touching or connected objects

(c) Apply an appropriate advanced segmentation method.

Method: Watershed Segmentation

(d) Justify how the method handles complexity.

Watershed segmentation:

- Separates touching objects.
- Uses gradient information instead of only intensity.
- Produces accurate boundaries for complex images.

(e) Provide a structured explanation.

Procedure:

1. Remove noise.
2. Find object markers.
3. Apply Watershed algorithm.
4. Obtain segmented regions.

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4. Broken Edges

(a) Interpret the scenario and explain why edges are broken.

After edge detection, some object boundaries appear incomplete due to noise or weak edge responses.

(b) Identify key issues affecting boundary continuity.

- Image noise
- Weak edges
- Low contrast
- Missing edge pixels

(c) Apply a suitable method to improve edge connectivity.

Method: Morphological Closing (Dilation followed by Erosion)

(d) Justify your approach logically.

Morphological closing:

- Connects broken edge segments.
- Fills small gaps.
- Produces continuous object boundaries.

- Preserves object shape.

(e) Present your explanation clearly.

1. Detect edges.
2. Apply dilation.
3. Apply erosion.
4. Obtain connected boundaries.

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5. Combined Requirement

(a) Interpret the scenario and define the combined requirement.

An industrial inspection system must remove image noise while preserving important object edges to detect defects accurately.

(b) Identify key issues involving noise and edge preservation.

- Random noise
- Loss of important edges
- False defect detection
- Reduced image quality

(c) Apply an appropriate sequence of techniques.

Technique Sequence:

1. Median Filter
2. Canny Edge Detection

(d) Justify the order and selection with strong reasoning.

- Median filter removes salt-and-pepper noise while preserving edges.
- Canny edge detection then accurately detects object boundaries.
- Performing edge detection before noise removal would produce many false edges.

(e) Present your response in a clear and structured format.

Flow:

1. Input image

2. Apply Median Filter
3. Remove noise